

SAN JUAN LA PUNTILLA, Puerto Rico

WMO: 994043

Lat: 18.459N

Lon: 66.116W

Elev: 5

StdP: 101.26

Time Zone: -4.00 (W04)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	22.2	22.8	N/A	N/A	N/A	N/A	N/A	N/A	14.6	26.1	12.2	25.9	2.4	220	0.449

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	4.1	31.3	N/A	30.7	N/A	30.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	5.0	140

Dehumidification DP/MCDB and HR

DP	HR	MCDB	1%			2%			0.4%			1%			Extreme Max WB	
			DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB		
			(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	(p)	(q)
10.5	9.4	8.5	DB	21.3	33.0	0.5	0.8	20.9	33.6	20.6	34.0	20.3	34.5	20.0	35.1	
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	26.8	25.2	25.2	25.2	26.0	27.0	27.9	27.9	28.2	28.3	28.1	27.0	26.0	
	DBStd	1.43	0.72	0.76	0.85	1.00	0.98	0.77	0.73	0.71	0.76	0.78	0.79	0.86	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDD10.0	6144	473	427	473	479	526	536	555	563	548	560	510	494	
	CDD18.3	3103	214	193	214	229	267	286	297	305	298	302	260	236	
	CDH23.3	31501	1573	1391	1523	1986	2676	3309	3501	3687	3597	3505	2687	2067	
Wind	WSAvg	4.4	4.4	4.6	4.5	4.8	4.3	4.6	5.2	4.9	3.7	3.6	3.6	4.3	
	PrecAvg	1736	104	59	75	130	154	146	151	198	185	191	186	166	
	PrecMax	2623	229	117	227	386	432	467	256	365	317	448	439	304	
	PrecMin	1088	15	28	15	26	25	18	74	78	67	101	65	10	
	PrecStd	332	51	23	50	91	104	92	53	80	81	111	115	85	
	DB	28.9	29.4	29.5	31.1	31.1	31.3	31.0	31.7	32.1	31.7	30.6	29.6		
	MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	27.9	28.2	28.2	29.5	30.2	30.6	30.2	30.6	31.2	31.0	29.7	28.7		
	2% MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% DB	27.4	27.5	27.5	28.5	29.5	30.0	29.8	30.1	30.6	30.4	29.4	28.2		
	5% MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10% DB	27.0	26.9	27.0	27.7	28.8	29.5	29.4	29.7	30.1	29.9	29.0	27.8		
	10% MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	0.4% MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2% MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
Mean Daily Temperature Range	MDBR	3.8	3.6	3.6	3.5	3.4	3.3	3.1	3.3	4.1	4.1	4.0	3.7		
	5% DB	4.2	4.4	4.2	4.6	4.3	3.9	3.5	3.6	4.6	4.5	4.4	4.1		
	5% MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
Clear-Sky Solar Irradiance	taub	0.371	0.386	0.386	0.413	0.453	0.527	0.525	0.488	0.442	0.391	0.385	0.377		
	taud	2.551	2.495	2.492	2.444	2.345	2.098	2.102	2.235	2.394	2.553	2.569	2.563		
	Ebn at Noon	914	915	924	895	848	780	783	818	858	899	892	896		
	Edh at Noon	97	107	111	117	128	162	162	142	121	100	95	93		
All-Sky Solar Radiation	RadAvg	4.50	5.24	5.80	6.09	5.86	5.99	6.06	5.91	5.58	5.00	4.39	4.22		
	RadStd	0.16	0.12	0.24	0.34	0.41	0.34	0.30	0.19	0.36	0.32	0.32	0.20		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	+0.28	+0.47	N/A	+0.33	N/A	N/A	N/A	N/A	+113	+112

Nomenclature: See separate page